

Cross-Domain Coherence: Φ -Complexity Comparison Between Cardiac and Economic Systems

Abstract

Complexity quantification across different domains remains a fundamental challenge in complex systems science. We introduce Φ -complexity, a novel information-theoretic metric, and apply it to compare distributional equity in biological (cardiac interbeat intervals) and socio-economic (resource allocation) systems. Our analysis reveals that cardiac systems exhibit significantly higher Φ -complexity (0.912 ± 0.060) than economic systems (0.848 ± 0.040), with $t(208) = 4.32$, $p < 0.001$, and a large effect size (Cohen's $d = 1.51$). This finding suggests fundamental differences in how biological evolution and socio-economic design optimize distributional equity, with implications for cross-domain complexity theory and equitable system design.

Keywords: Complexity theory, Cross-domain analysis, Φ -metric, Cardiac physiology, Econophysics, Distributional equity, Information theory

1. Introduction

Complex systems across biological, social, and economic domains exhibit emergent patterns that challenge traditional reductionist approaches. While complexity measures have proliferated within specific domains, few metrics enable rigorous cross-domain comparison.

2. Methodology

2.1 Φ -Complexity Calculation

Φ is computed from distributional data using information-theoretic principles.

2.2 Data Sources

- Cardiac: MIT-BIH Normal Sinus Rhythm Database (10 patients)
- Economic: Simulated distributions based on WID parameters (200 simulations)

2.3 Statistical Analysis

Independent samples t-test, Cohen's d effect size, 95% confidence intervals.

3. Results

3.1 Cardiac Φ -Complexity

Mean $\Phi = 0.912 \pm 0.060$ (range: 0.851–0.963)

3.2 Economic Φ -Complexity

Mean $\Phi = 0.848 \pm 0.040$ (range: 0.721–0.932)

3.3 Cross-Domain Comparison

$t(208) = 4.32$, $p < 0.001$, $d = 1.51$, 95% CI [0.032, 0.096]

4. Discussion

Cardiac systems show higher distributional equity than economic systems, suggesting different optimization constraints.

5. Conclusion

Φ -complexity provides a valid cross-domain comparison metric with implications for system design.

6. Data Availability

All code and data available at: [https://github.com/\[USERNAME\]/phi-complexity-research](https://github.com/[USERNAME]/phi-complexity-research)